Homework 3

Control Systems Introduction (Chapter 1)

1. Read Chapter 1 of the textbook. Briefly define the following terms:
    a. Controller
    b. Actuator
    c. Process
    d. Plant
    e. Sensor
2. Draw a block diagram with labeled blocks using the terms in the previous problem. Also, label the following signals: reference, control signal, disturbance, output, sensor noise, sensor reading, and error (the difference between the reference signal and sensor reading). Hint: Figure 1.2 of the textbook has most of these.
3. What is the difference between an open-loop control and closed-loop control? Briefly explain when you would use each one, and give a real-world example of each (that was not discussed in class).
4. Typical control systems have requirements on stability, tracking, disturbance rejection, and robustness. Briefly explain each one (hint: Chapter 1 discusses these).

Laplace Transforms (Section 3.1)

5. Book problem 3.3 (use a Laplace table)
6. Book problem 3.7(a) by hand (can use a Laplace table)
7. Book problem 3.7(d) using Matlab. Hint: use ilaplace()
8. Book problem 3.9(a)

System Identification

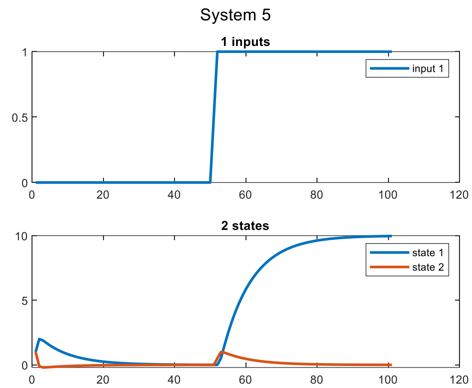
Identifying and modeling dynamic systems is often a prerequisite to being able to apply control. In this case study, you will create models for 10 dynamic systems. Download file "RunExperiment.p" from Canvas, and make sure it is in your working directory. Understand the inputs and outputs of the function, listed below:

```
>> help RunExperiment
 RunExperiment(id, U, make_plot):  Runs a system simulation for input matrix U
    inputs:  id: integer from 1 to 10 that sets what system to simulate
             U:  [Nt x Nu] input matrix, where Nt is the number of
             timepoints and Nu is the number of inputs for the system.
             Each column is a input signal.
             make_plot: Boolean for if a plot should be generated
    outputs: X:  [Nt x Nx] output matrix, where Nt is the number of
             timepoints and Nx is the number of states.  Each column is how that
             state evolves with time.
```

For example, to run system 5 with a step input, we could write something like

```
t = (0:.1:10)';
U = ones(length(t), 1).*heaviside(t-5);
X = RunExperiment(5, U, true);
```

Note that the initial conditions of the model cannot be modified – all states start at a value of 1.



9. Describe why having a good system model could help with designing a control scheme.
10. Based on your prior classes and laboratory experiments, describe common ways for identifying systems. Hint: think impulse response, step response, frequency sweeps, etc.
11. For each of the 10 systems, attempt to create a system model. You should attempt to create a recursive model, such that the state of the system at the next time step is a function of the states and inputs of the system at prior timesteps. The model may be nonlinear.

$$X[k+1] = f(X[k], X[k-1], \dots, U[k], U[k-1], \dots.)$$

Explain what experiments you ran (what input signals U) to make the system model.

Note that this can be very hard in some cases! You are not expected to get every system, or be able to come up with a good model. Try to get at least 7/10 systems.

12. For each of the 10 systems, plot your system model output and compare it to the true system output. How close did you get?
13. Complete the summary table below. "Model class" refers to a generic classification of the system type based on your experiments. Examples include underdamped mass spring dampers, heat diffusion, time delay, smoothing filters, integrators, etc.

System	# inputs	# outputs	Model	Model class
1	1	1	x[k+1] = -x[k]	Undamped oscillations indifferent to input
2	1			
3	2			
4	1			
5	1			
6	1			
7	1			
8	1			
9	1			
10	1			

Optional Bonus Problems

14. Solve book problem 3.9(a) in Matlab using Laplace transforms.
Hint: (some of the code is blacked out for you to fill in)

```
%% Tested with Matlab 2018a

syms y(t) real
syms t real
syms s
syms Y

y0 = [redacted];
dy0 = [redacted];

dy = diff(y,t,1)
dy2 = [redacted]

eqn = dy2+ dy+ 3*y - 0

eqn2 = laplace([redacted], t, s)

eqn3 = subs(eqn2,str2sym('y(0)'), y0)
eqn3 = subs(eqn3,str2sym('[redacted]'), dy0)
eqn3 = subs(eqn3,[redacted], Y)  % This may change based on your version of Matlab

eqn4 = solve(eqn3, Y)

y = ilaplace(eqn4, t)
pretty(y)

y = vpa(y,4)
```

The code produces the following output (may vary slightly depending on Matlab version and your computer configuration):

```
dy(t) =

diff(y(t), t)


dy2(t) =

diff(y(t), t, t)


eqn(t) =

3*y(t) + diff(y(t), t) + diff(y(t), t, t)


eqn2 =

s*laplace(y(t), t, s) - y(0) - s*y(0) + s^2*laplace(y(t), t, s) - subs(diff(y(t), t), t, 0) + 3*laplace(y(t), t, s)


eqn3 =

s*laplace(y(t), t, s) - s + s^2*laplace(y(t), t, s) - subs(diff(y(t), t), t, 0) + 3*laplace(y(t), t, s) - 1


eqn3 =

s*laplace(y(t), t, s) - s + s^2*laplace(y(t), t, s) + 3*laplace(y(t), t, s) - 3


eqn3 =
```

```
3*Y - s + Y*s + Y*s^2 - 3


eqn4 =

(s + 3)/(s^2 + s + 3)


y =

exp(-t/2)*(cos((11^(1/2)*t)/2) + (5*11^(1/2)*sin((11^(1/2)*t)/2))/11)

       /                     / sqrt(11) t \  \
       |              sqrt(11) sin| ---------- | 5 |
  /  t \ |   / sqrt(11) t \           \    2    /  |
exp| - - | | cos| ---------- | + ---------------------------- |
  \  2 / \   \    2    /           11          /


y =

exp(-0.5*t)*(cos(1.658*t) + 1.508*sin(1.658*t))

>>
```

15. Use your code to solve 3.9(f).